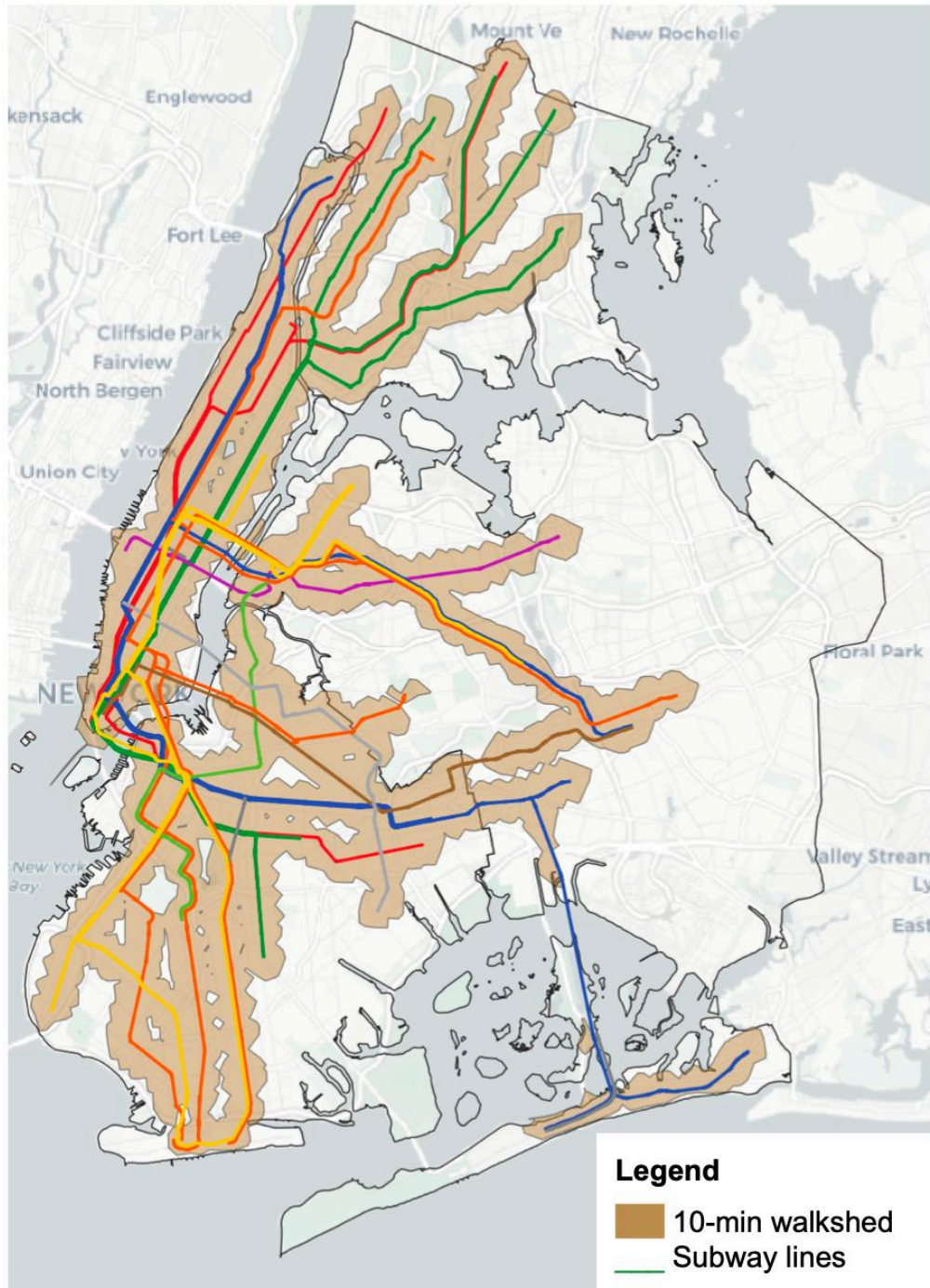


Found Visualizations assignment

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10 Minute Trip — KPF Urban Interface

Area Walkable Within 10 Minutes from New York City Subway Stations



<https://ui.kpf.com/blog/2017/12/13/ten-minute-trip?utm.com>

"10 Minute Trip" is a data visualization and urban analysis tool developed by KPF Urban Interface in 2017. The project explores the relationship between time, distance, and accessibility in New York City. Rather than representing the city simply through geographic distance, the visualization asks what a person can actually reach within ten minutes. It combines walking and subway travel to create a different understanding of urban space. Starting from a selected location, the visualization identifies buildings that can be reached within ten minutes and uses color shading to represent travel time in one-minute increments. Importantly, the analysis does not simply draw a ten-minute walking radius. It considers subway stations, available train lines, average waiting times, additional stops, transfers, walking to and from stations, and the final destination. This makes the visualization an attempt to represent the city as a network of movement rather than simply a physical map.

The scope of the project is New York City, and KPF applies the tool at several different scales. Individual transit locations such as Grand Central, Queensboro Plaza, Atlantic Avenue/Barclays Center, Sunnyside, and Red Hook are compared, while another analysis applies the visualization along the entire length of Broadway, from 220th Street to Battery Park. These comparisons reveal major differences in urban connectivity. For example, KPF's analysis shows that Grand Central has 4,157 accessible buildings within the ten-minute analysis, while Queensboro Plaza has 5,024. Atlantic Avenue/Barclays Center has 7,660 accessible buildings. The project also compares less connected areas such as Red Hook and Sunnyside, demonstrating how neighborhoods with limited subway access can become relatively isolated from the rest of the city.

The visualization was created by KPF Urban Interface, a group within the architecture firm Kohn Pedersen Fox that focuses on urban analytics, computational design, and technology. KPF Urban Interface describes its work as using data collection, scenario analysis, and visualization to better understand and design contemporary cities. This context is important because "10 Minute Trip" is not intended only as a graphic for communicating information. KPF presents the tool as a potential design and analysis platform for architects, planners, urban designers, and urbanists.

I find the visualization generally credible because its method reflects an important reality of New York: physical distance and travel time are not necessarily the same thing. A destination that is geographically farther away can sometimes be reached more quickly because of a direct subway connection, while a location that appears nearby can take longer because of walking routes, transfers, or poor transit connections. KPF explicitly explains the different components included in its calculations, which makes the methodology understandable. However, the visualization also has limitations. It represents accessibility as a calculated travel-time model rather than a complete representation of lived experience. It does not fully capture variables such as delays, overcrowding, weather, individual walking speed, accessibility for people with disabilities, or how safe or comfortable a person feels during a journey. The original project explains its methodology, but the data sources and assumptions could be documented more transparently. Therefore, I would consider it credible as a model of urban connectivity, but not as a perfect representation of every person's actual experience.

Visually, the project is effective because the map is an appropriate form for spatial data. The color gradients immediately communicate how travel time expands from a starting point,

while the buildings provide a physical representation of what is accessible. The hierarchy is relatively clear: the starting location and transit network establish the context, while the colored buildings communicate the main information. The visualization is particularly successful because it turns time into a spatial quality. Instead of reading “seven minutes” as an isolated number, the viewer can see what seven minutes actually means within the physical city.

The most interesting intellectual contribution of “10 Minute Trip” is that it challenges the conventional idea of a map. A traditional map tells us where things are located, but this visualization tells us what is reachable. KPF describes the relationship between time and distance as fluid and shows how transit can effectively “collapse” or “expand” urban space. This makes the visualization compelling because it reveals an invisible urban system: connectivity. The city is no longer represented only as buildings, streets, and blocks, but as a network of possible movements between places.

There are several opportunities to develop the project further. The same concept could be used to create five-, ten-, fifteen-, or thirty-minute maps and compare how accessibility changes over time. The visualization could also distinguish between different types of destinations, such as schools, hospitals, parks, grocery stores, cultural institutions, or workplaces. Most importantly, I would add more human-centered information. The current visualization asks, “What can someone reach within ten minutes?” A future version could ask, “Who can reach what within ten minutes, and under what conditions?” Adding pedestrian counts, accessibility information, demographic data, building uses, and actual travel-time data could create a much richer understanding of urban experience.

What I find most useful about “10 Minute Trip” for architecture is its ability to treat data visualization as a design tool rather than simply an illustration. It demonstrates that movement can be used to understand the city just as buildings, streets, density, and zoning can. The project makes an invisible system visible and shows how transportation infrastructure affects the way people experience urban space. Ultimately, “10 Minute Trip” suggests that the city should not only be understood as a collection of physical places, but also as a constantly changing network of relationships between people, buildings, transportation, and time.